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Regenerative Braking Control with Bidirectional Converter for EV Applications

Hüseyin Kurt *

[1]*Department of Electrical Electronics Engineering, Istanbul Aydin University, Istanbul, Turkey*

**huseyinkurt1@aydin.edu.tr*

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Abstract – Regenerative braking is a key technology in electric vehicles (EVs) for improving energy efficiency and extending driving range by converting kinetic energy into stored electrical energy during deceleration. This study proposes a dual-mode regenerative braking control strategy for a permanent magnet synchronous motor (PMSM) drivetrain, integrating a bidirectional buck–boost DC–DC converter with a full-bridge inverter. Unlike conventional unidirectional converter systems, the proposed architecture enables bidirectional power flow between the motor and the battery, allowing efficient energy recovery and controlled deceleration under a wide range of operating conditions. A field-oriented control (FOC) scheme is employed for precise torque and flux regulation, while an intelligent supervisory logic dynamically selects the converter mode based on real-time motor torque feedback. The complete system, including realistic battery dynamics, is modeled and simulated in MATLAB/Simulink under flat-road, low-speed driving scenarios, where conventional regenerative methods often exhibit reduced performance. Simulation results demonstrate that the proposed approach increases recovered energy by more than 22% compared to a baseline unidirectional converter, while maintaining DC bus voltage stability, suppressing torque ripple, and ensuring smooth mode transitions between acceleration and regenerative braking. These findings validate the practicality of the proposed strategy, indicating its potential for real-world implementation to enhance battery state-of-charge (SOC) management, extend EV range, and improve overall drivetrain energy efficiency without adding significant system complexity.

Keywords – Regenerative Braking Control, Bidirectional Buck-Boost Converter, Field-Oriented Control, SVPWM, PMSM, Electrical Vehicle

I. INTRODUCTION

The rising adoption of electric vehicles (EVs) has placed increasing importance on energy efficiency and battery longevity [1, 2]. Regenerative braking (RB) has emerged as a key technique for improving energy utilization by recovering kinetic energy during deceleration and storing it back into the battery system [3,4]. However, conventional RB methods are often limited in low-speed scenarios due to inadequate back-EMF generation [5, 6]. Permanent Magnet Synchronous Motors (PMSMs) are adopted in EV applications due to their power density, high efficiency, and superior torque characteristics [7, 8]. However, the control of PMSMs during regenerative braking introduces several technical challenges, particularly when considering bidirectional energy flow between the battery and motor under dynamic driving conditions [9, 10]. These challenges become more pronounced when the system is expected to switch seamlessly between motoring and regenerative modes without compromising performance or stability [11]. In conventional regenerative systems, a unidirectional control strategy is often employed, which limits the energy recovery potential and fails to optimize the charging dynamics of the battery across road scenarios [12, 13]. To address these limitations, this study proposes a regenerative braking system based on a bidirectional cascaded DC-DC converter included with a full-bridge inverter [14]. The dual-mode architecture enables two distinct operating strategies: (1) Boost mode during braking, allowing energy recovery by elevating the DC-link voltage for effective battery charging [15] and (2) Buck mode during acceleration, facilitating controlled battery discharge for propulsion [16]. A Field-Oriented Control (FOC) strategy is integrated to ensure precise torque and flux regulation of the PMSM [17].

II. MATERIALS AND METHOD

This section presents the modeling, design, and control approach for the regenerative braking system for a PMSM-based electric vehicle at low speed. The MATLAB Simulink simulation model consists of a bidirectional DC-DC converter, an SVPWM inverter for PMSM drive, and FOC algorithm.

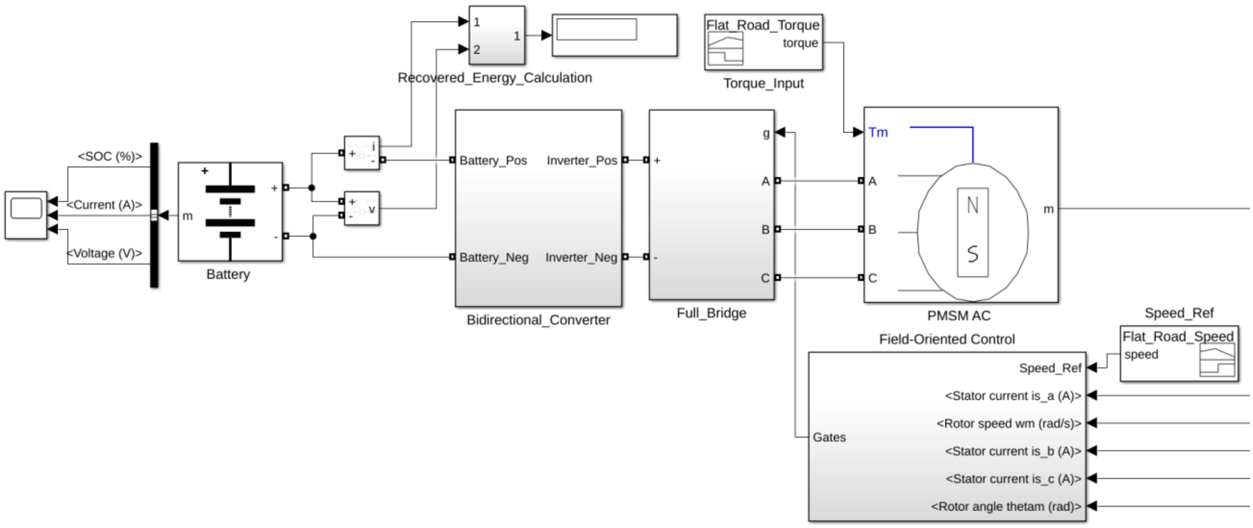

Fig. 1 Regenerative braking model blocks

The components of simulation are; a lithium-ion battery pack serving as the DC energy source and sink, a bidirectional DC-DC converter capable of operating in both buck (regenerating) mode and boost (motoring) mode, a full-bridge inverter for AC excitation of the PMSM, a PMSM as the traction motor.

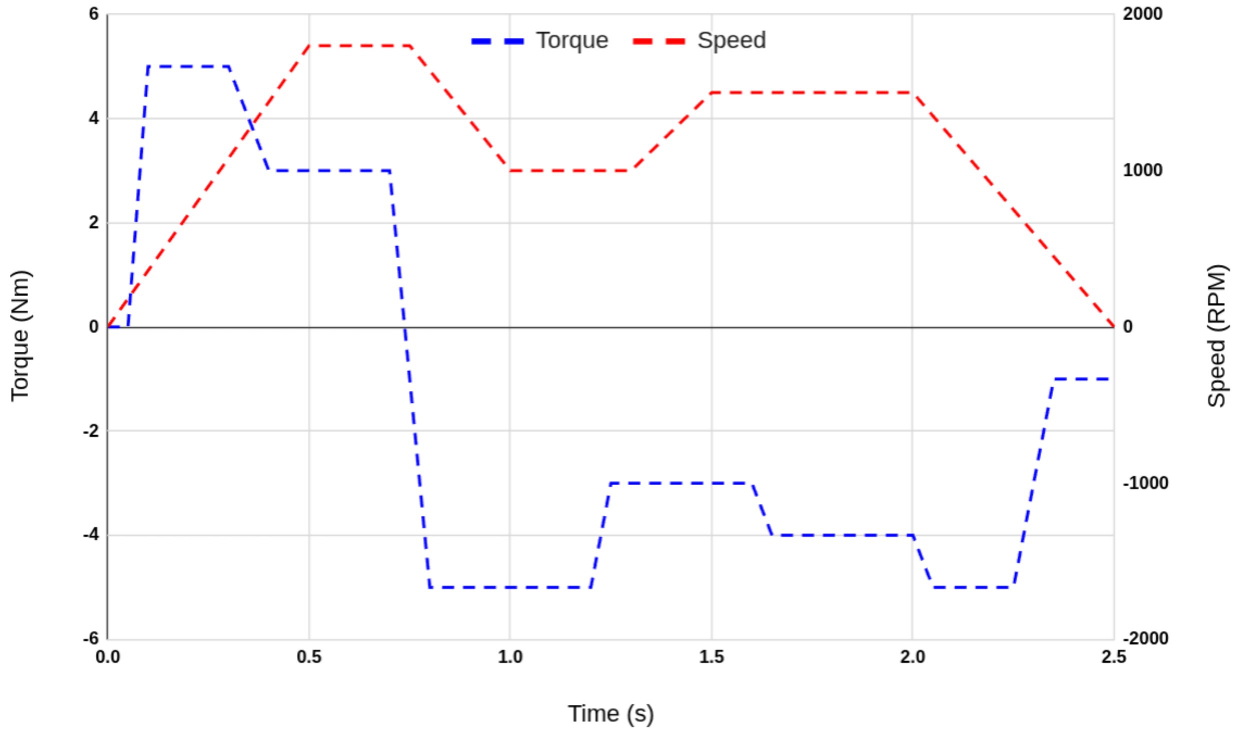

Fig. 2 Road torque and reference speed loaded to the PMSM motor

The reference speed values presented in figure 1 are specified for low-speed operation (≤ 20 km/h), while the torque value is determined under a flat road profile condition.

$$Speed\left(\frac{km}{h}\right) = \frac{Motor\ RPM}{Gear\ Ratio} \times \pi \times D_{wheel} \times \frac{60}{1000} \tag{1}$$

Where:

- $Motor\ RPM$ = Motor speed in revolutions per minute
- $Gear\ Ratio$ = Total reduction (motor turns / wheel turns)
- D_{wheel} = Wheel diameter in meters
- $\frac{60}{1000}$ = Converts meters/min to km/h

According to Equation 1, the simulation reference speed ranges from 0 km/h to 20.36 km/h (corresponding to 0 RPM to 1800 RPM).

The PMSM is controlled using the Field-Oriented Control (FOC) technique for both motoring and regenerative modes. The FOC algorithm consists of; Clarke and Park transformations to convert stator currents from three-phase to d-q rotating reference frame, PI controllers for torque and flux control via i_d and i_q currents, Space Vector PWM (SVPWM) for inverter switching pattern generation.

Table 1. Parameters of the model

Parameter	Value
Motor nominal voltage	300 Vdc
PMSM rated power	2830 W
Converter capacitance	1000 µF
Converter inductance	1 mH
Converter switching frequency	5 kHz
Battery nominal voltage	280 V
Rated capacity (Ah)	12 Ah
Wheel radius	0.3 m
Gear ratio	10

The simulated model parameters are presented in Table 1. The rated values of the selected motor in Simulink are 6 Nm, 300 Vdc, and 4500 RPM. The vehicle has a wheel radius of 0.3 m and a total mass of 1200 kg.

III. RESULTS

The proposed model was simulated for a duration of 2.5 seconds. A key innovation of this study is the implementation of a converter control strategy governed by real-time motor torque. The operating modes are defined as follows:

If $T_e > 0$: Motoring phase → Boost mode,

If $T_e < 0$: Regenerative braking phase → Buck mode,

where T_e is motor electrical torque.

Buck mode refers to the regenerative phase in which energy flows from the DC link (high voltage) to the battery (low voltage) to charge the battery during braking. The battery voltage, as well as the converter capacitance (C) and inductance (L) values, are calculated for buck mode as follows:

$$V_B = V_{DC} \times D \tag{2}$$

$$C = \frac{(1-D)V_B}{8L\Delta V_B f} \tag{3}$$

$$L = \frac{D(V_{DC} - V_B)}{\Delta I_L f} \tag{4}$$

Boost mode corresponds to the motoring phase, in which energy flows from the battery to the motor. The DC link voltage, capacitance (C), and inductance (L) values for this mode are calculated as follows:

$$V_{DC} = \frac{V_B}{(1-D)} \tag{5}$$

$$C = \frac{D \times V_{DC}}{R_{out}\Delta V_{DC} f} \tag{6}$$

$$L = \frac{V_B \times D}{\Delta I_L f} \tag{7}$$

The inductance (L) and capacitance (C) are calculated separately for both buck and boost modes. The L and C values of the bidirectional converter are then determined by selecting the maximum calculated values from these two modes. In this study, the key performance indicators (KPIs) monitored include the battery state of charge (SOC), recovered energy, and the system's speed and torque responses.

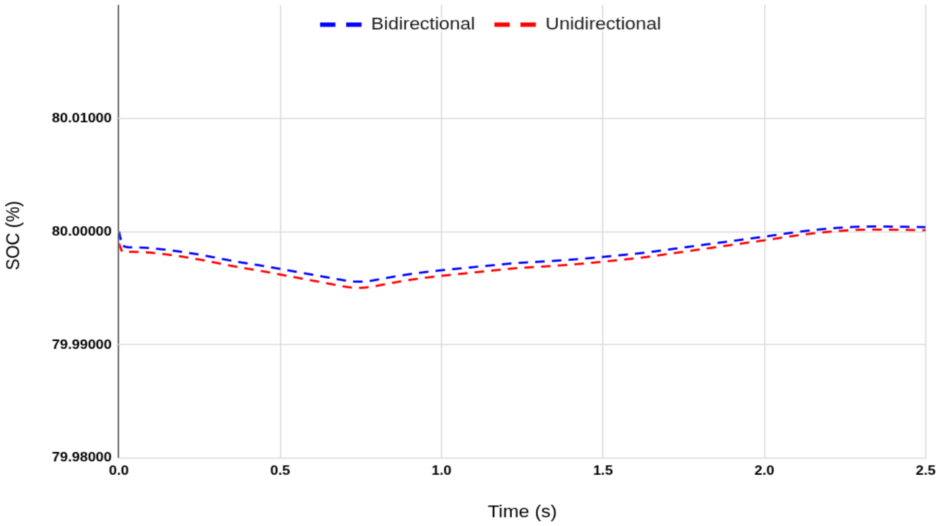

Fig. 3 The battery state-of-charge for both converters

Both converters are capable of regenerating energy under negative torque; however, the bidirectional converter is significantly more effective than the unidirectional converter in retaining this energy.

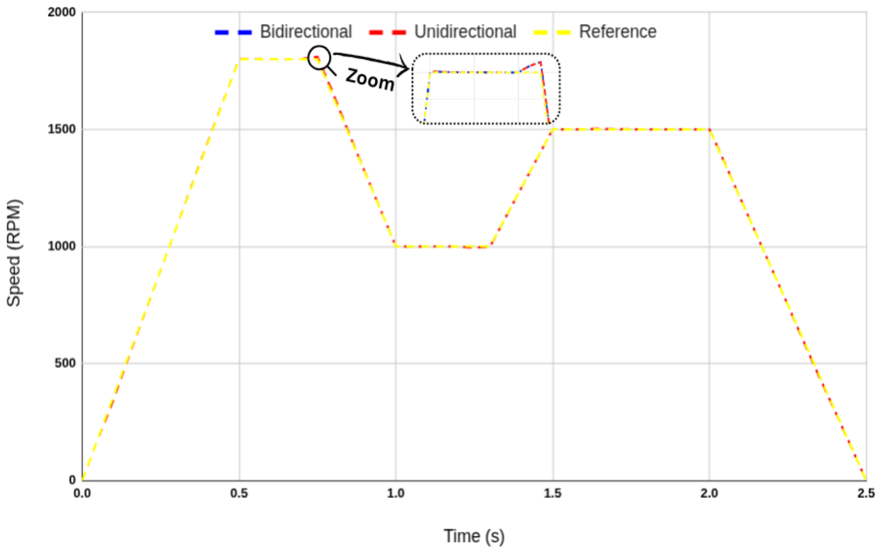

Fig. 4 Reference speed and unidirectional and bidirectional converter speed responses

Figure 4 illustrates the speed response of the reference and compared converters under the applied road profile. In both cases, the speed tracking performance is satisfactory, as each converter employs a field-oriented control (FOC) strategy combined with a proportional–integral (PI) controller.

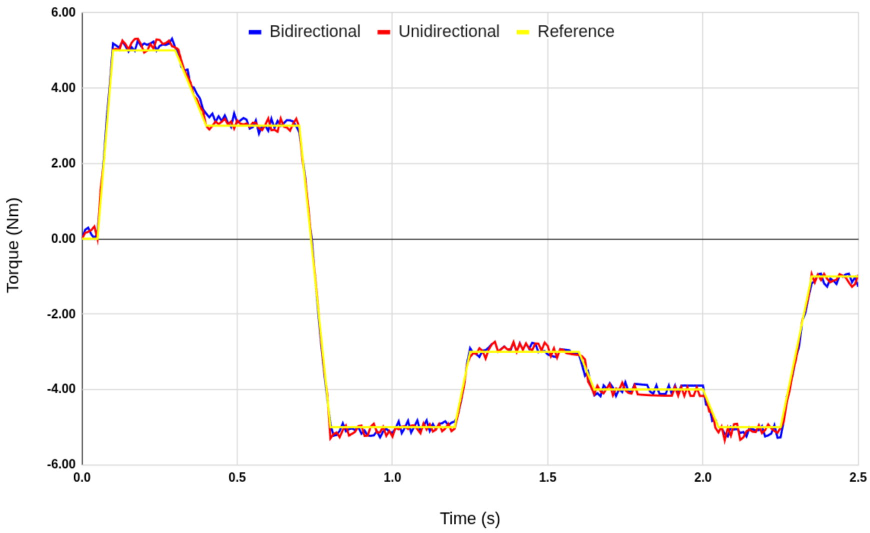

Fig. 5 The torque responses for both converter with reference torque

Figure 5 presents the torque response of the two converter types under the applied load profile. During braking, the torque becomes negative, activating boost mode and initiating energy recovery. The smooth transition between acceleration and regenerative braking phases confirms the effectiveness of the dual-mode operation.

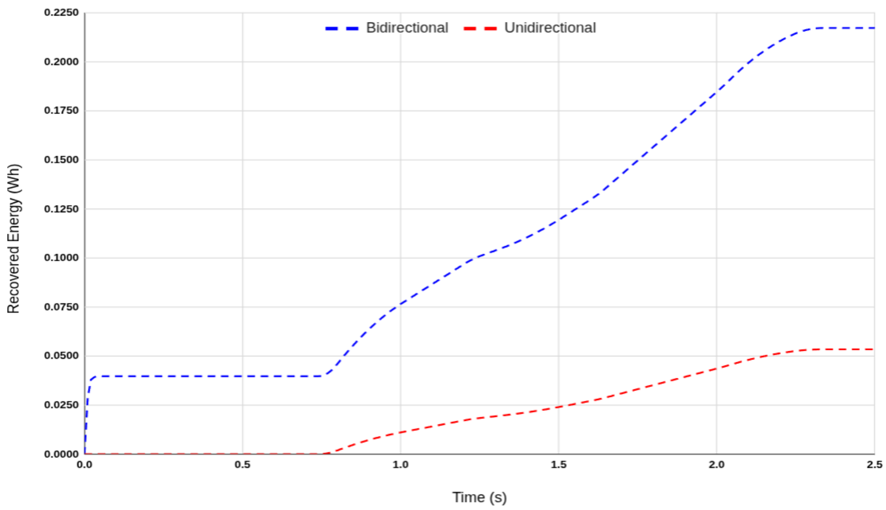

Fig. 6 Recovered energy for two converters

The performance of the proposed bidirectional converter strategy is compared against that of a baseline unidirectional converter.

Table 2. Recovered Energy

Converter Type	Average Regenerative Power (W)	Total Energy Recovered (Wh)
Bidirectional	312.91 W	0.2173 Wh
Unidirectional	256.46 W	0.1781 Wh

Over a simulation period of 2.5 seconds, the bidirectional converter recovered approximately 0.2173 Wh of energy, compared to 0.1781 Wh recovered by the unidirectional converter. This corresponds to an increase of about 22.01% in recovered energy, demonstrating the superior regenerative capability of the proposed bidirectional converter. These findings indicate the effectiveness of the bidirectional converter for regenerative control strategy. The intelligent mode-switching logic enables efficient energy flow management and ensures seamless operation without introducing torque ripple or system instability. Overall, the results indicate strong potential for real-world implementation, with expected improvements in electric vehicle (EV) range and overall energy efficiency.

IV. CONCLUSION

This study has presented a dual-mode regenerative braking control strategy for a PMSM-based electric vehicle drivetrain, employing a bidirectional cascaded DC–DC converter. By integrating this converter topology with a field-oriented control (FOC) scheme, the system achieves efficient bidirectional power flow management between the motor and the battery. Simulation results, validated under flat-road, low-speed driving scenarios, demonstrated that the proposed approach offers superior energy recovery performance. Compared to conventional unidirectional converter-based regenerative braking systems, the bidirectional converter strategy increased recovered energy by more than 22.01%, while ensuring system stability, effective DC bus voltage regulation, and smooth torque transitions. The successful implementation of the dual-mode converter logic in MATLAB/Simulink confirms the practicality of this architecture for real-world electric vehicle applications. Beyond enhancing energy efficiency and battery state-of-charge (SOC) management, this control strategy has the potential to extend the operational range of electric vehicles without introducing significant system complexity, thereby offering a promising solution for next-generation EV drivetrain designs.

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